



CHEMISTRY Department
SIMHADRI



Ref: SMTTP/Chemistry/2022-23/COAL LMI/746909

Date: 09/01/2023(16:20:52)

Subject: LMI on Sampling and Analysis of Coal - Revised

LMI doc no. LMI- CHEM/OPS-26 / REVISION 4, "LMI ON SAMPLING AND ANALYSIS OF COAL" has been revised in line with COS-ISO-00-OGN/OPS/CHEM/ 006 Rev. No.: 5 Nov. 2022.

The following changes are being done in the revised LMI :

- Detailed description of test methods has been deleted and a summary table have been incorporated regarding the standard test methods.
- Empanelment of third-party sampling agencies at mine end and unloading end as per latest circular by MOP included.
- Imported Coal Sampling modalities have been changed as per the latest guidelines.
- Coal as fired sampling Section and disposal frequency has been included.
- General guidelines included in Annexure-1 for isolation of OHE for wagon top sampling.

The document enclosed for kind perusal and approval please.

Submitted by:

सुनीता गुलीपिल्ली - प्रबंधक

SUNEETA GULIPILLI - 101282 (MANAGER ,CHEMISTRY) ,Simhadri
(SUNEETAGULIPILLI@NTPC.CO.IN, 9493547082)

On: 09/01/2023(16:20:52) **Ref:** SMTTP/Chemistry/2022-23/COAL LMI/746909

May please be approved

Submitted by:

दीपशिखा वर्मा - उप महाप्रबंधक (रसायन)

Deepshikha Verma - 007773 (DEPUTY GENERAL MANAGER (CHEMIST) ,CHEMISTRY)
(DEEPSHIKHAVERMA@NTPC.CO.IN, 9650995244) ,Simhadri

On:10/01/2023(17:03:05) **Ref:**SMTTP/Chemistry/2022-23/COAL LMI/746909

Submitted by:

पीयूष श्रीवास्तव - अपर महाप्रबंधक

Piyush Srivastava - 006316 (ADDL. GENERAL MANAGER ,CHEMISTRY)
(PIYUSHSRIVASTAVA@NTPC.CO.IN, 9650999535) ,Simhadri

On:10/01/2023(19:32:56) **Ref:**SMTTP/Chemistry/2022-23/COAL LMI/746909

Submitted by:

Amulya Kumar Behera - 097433 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(AMULYABEHERA@NTPC.CO.IN, 9425534368) ,Simhadri

On:12/01/2023(10:53:16) **Ref:**SMTTP/Chemistry/2022-23/COAL LMI/746909

Submitted by:

सन्जित मजूमदार - अपर महाप्रबंधक

Sanjit Majumdar - 005065 (ADDL. GENERAL MANAGER ,FUEL MANAGEMENT)
(SMAZUMDAR@NTPC.CO.IN, 9650990065) ,Simhadri

On:13/01/2023(09:51:10) **Ref:**SMTPP/Chemistry/2022-23/COAL LMI/746909

Submitted by:

Vishwa Mohan Choudhary - 003932 (GENERAL MANAGER ,FUEL MANAGEMENT)
(VMCHOUDHURY@NTPC.CO.IN, 7588691232) ,Simhadri

On:16/01/2023(11:56:18) **Ref:**SMTPP/Chemistry/2022-23/COAL LMI/746909

May please be approved

Submitted by:

Dhurjati Prasad Patra - 004573 (GENERAL MANAGER ,O & M)
(DPPATRA@NTPC.CO.IN, 8275045446) ,Simhadri

On:17/01/2023(10:36:46) **Ref:**SMTPP/Chemistry/2022-23/COAL LMI/746909

Approved & Submitted by:

संजय कुमार सिन्हा - महाप्रबंधक

Sanjay Kumar Sinha - 005185 (GENERAL MANAGER ,HEAD OF PROJECT)
(SKSINHA06@NTPC.CO.IN, 9431215239) ,Simhadri

On:17/01/2023(15:28:12) **Ref:**SMTPP/Chemistry/2022-23/COAL LMI/746909

NTPC
SIMHADRI/

LMI- CHEM/OPS-26 / REVISION 4

LMI ON SAMPLING AND ANALYSIS OF COAL



Approved for implementation by GGM (Simhadri)

Sampling and Analysis of Coal

Contents

S.No.	Description	Page Nos.
1	Introduction	3
2	Superseded Documents	3
3.1	Scope	3
3.2	Responsibility	4
4.1	Coal as fired sampling & preparation	5
4.2	Domestic coal sampling & preparation at mines	6
4.3	Collection of samples from wagons	8
4.4	Collection of samples of coal dispatches by road	10
4.5	Stack yard coal sampling	11
4.6	Preparation of domestic coal samples	11
4.7	Preparation of total moisture sample and determination of total moisture	12
4.8	Modalities for imported coal quality determination	12
4.9	Imported coal - sampling guidelines for NTPC Simhadri station end sample collection:	13
5.0	Corrections on quality variations	15
5.1	Gross calorific value-based grading system	16
5.2	Summary of test methods used for coal samples analysis	17
5.3	Reporting of results	17
5.4	Criteria for rejection of suspect test results	18
5.5	Associated documents/References	19

1.0 INTRODUCTION

In coal fired power stations, coal is the most important input for commercial use, also for management of proper combustion and operational efficiency of the boiler. A good sampling system is one that collects a representative sample and from whole quantity of the coal in consideration. For which minimum specified number of increments of specified weight, at specified interval shall be collected.

Correct quality evaluation of coal depends on its representative sampling, preparation and analysis. Accurate analysis and reporting of results is essential to evaluate the original coal quality and the purpose for which it is intended.

2.0 SUPERSEDED DOCUMENT

NTPC-SIMHADRI/LMI- CHEM/OPS-26 / REVISION 3

3.0 SCOPE & RESPONSIBILITY

3.1 Scope: This LMI provides guidelines for sampling and testing of coal as fired, Domestic coal, imported coal, coal from stock piles. Also, provides standards that are being in use for sampling, preparation, testing processes.

3.2 Responsibility for procedures -

Sl no	Activity	Responsibility
1.	Ensuring collection and preparation of coal as fired samples as per procedure in Section 4.1	HOD (Chemistry)
2.	Ensuring that Samples of domestic coal are collected by the CIMFR either manually or through any suitable mechanical sampling at loading end and unloading ends as per section 4.2	HOD (Chemistry)
3.	Ensuring collection of domestic coal samples from wagons by as per section 4.3	HOD – FM & HOD (Chemistry)
4.	Collection of coal samples dispatch by road mode 4.4	HOD – FM & HOD (Chemistry)
5.	Ensuring collection of samples from stock yard piles as per section 4.5	HOD – FM & HOD (Chemistry)
6.	Preparation of domestic coal samples as per 4.6	HOD (Chemistry)
7.	Preparation of Total moisture sample and determination of total moisture as per 4.7	HOD (Chemistry)
8.	Modalities for imported coal quality determination as per 4.8	HOD (Chemistry)
9.	Imported coal - sampling guidelines for NTPC Simhadri station end sample collection 4.9	HOD – FM & HOD (Chemistry)
10.	Corrections based on quality variations as per 5.0	HOD – FM & HOD (Chemistry)

4.0 COAL SAMPLING & PREPARATION:

4.1 COAL AS FIRED SAMPLING & PREPARATION:

Coal as fired sampling results required for the internal monitoring of the impacts of Coal quality related to combustion characteristics and efficiency issues and to ascertain the blending ratio for the stations where blended coal is fired. Coal received in power station is crushed and fed to bunkers, coal sample collected before bunkering / firing is termed as coal as fired sample. In order to take representative coal sample of coal as fired the sampling point should be safe, easily accessible and convenient for collecting the sample.

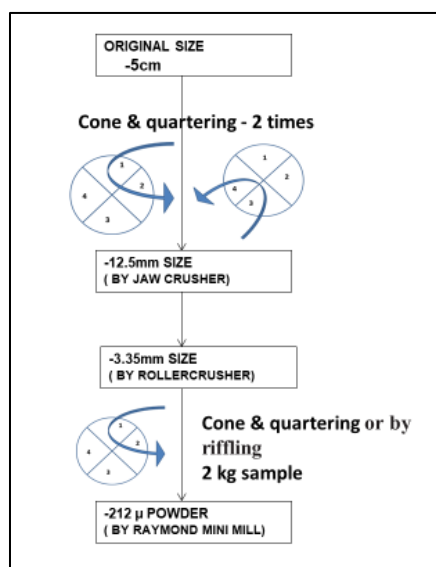
For each PAT assessment, (M&V year) coal as fired sampling and testing is to be carried out as per issued guidelines for PAT-II cycle.

4.1.1 COLLECTION OF SAMPLES FROM FEEDERS:

- a) Coal as fired Sample can be collected manually from sampling ports at the individual gravimetric /volumetric feeder, from the bunker to individual mills at equal interval in all three shifts.
- b) Enough coal quantity not less than 450 kgs/day is collected from the feeders.
- c) The sample shall be collected in 5 number of increments with 4 kg sample per increment.

4.1.2 PREPARATION OF SAMPLE:

- a) The gross sample of the day shall be mixed and crushed to (-) 12.5 mm by a jaw crusher and finally to 3.35 mm by a Palmac type of reduction mill.



Picture - 1

- b) The crushed materials shall be reduced by using a coal sample riffler or manual cone and quartering reduction till 2 kg.
- c) Finally, the above 3.35 mm size sample for analysis purpose shall be reduced to 212 micron size by pulverizer. (The process depicted in Picture -1).
- d) Excessive grinding of sample shall be avoided as it is harmful.
- e) When imported coal is fired, Coal as fired sample for individual feeder is carried out on weekly basis.
- f) During part load operation, VM % of Feeder Coal needs to be monitored.
- g) The testing frequency is on daily basis and reporting shall be on stagewise as per the table given in picture – 2 (Reference: COS-ISO-00-OGN/OPS/CHEM/015 -Code of Practice on Power Plant Chemistry).

S.N.	DATE OF SAMPLING	QTY (MT)	TOTAL MOISTURE (%)	ASH (%)	VM (%)	FC (%)	GCV (Kcal/kg)	REMARKS

Table 1

4.2 Third party Coal sampling modalities and Sample Preparation:

The Third-party coal sampling procedure has been jointly agreed between NTPC and coal supplier and CIMFR and form part of Fuel supply agreement. The procedures as described are based on FSA / BIS standards.

In case Third party coal sampling agreement (Tripartite) among supplier (CIL), NTPC and CIMFR is in place then at loading end sampling will be done by CIMFR as per the signed agreement. Collection of samples at unloading point will be done by CIMFR as per bilateral agreement.

4.2.1 Coal sampling procedure (CIMFR based third party sampling):

- a) The detailed modalities for Third Party sampling procedures which have been jointly agreed between NTPC and coal supplier and form part of Fuel supply agreement (Schedule V)/ relevant IS 436-part-I, Sec-I, 1964 specification or its latest version are to be followed for coal sample preparation and analysis, for determination of coal quality for payment purpose. (Note: The procedure may change at time of review in Fuel Supply Agreement).
- b) Samples of Coal shall be collected by the Third Party either manually or through any suitable mechanical sampling arrangement including Auger Sampling method if physically operational, at each of the Delivery Points (LOADING POINT) for determining the quality of Coal in presence of representatives of NTPC and Coal India.

- c) NTPC-Simhadri being a rail fed station, for the purpose of sampling, each rake (source wise, grade wise, plant wise and siding wise) of Coal supplied from one Delivery Point shall be considered as a lot.
- d) Sample for determination of Total Moisture shall be segregated from the sample collected at the Delivery Point by the Third Party, and prepared and analysed, as per the BIS standards.
- e) The Gross Sample collected as mentioned above will be for determination of Moisture, Ash & GCV on equilibrated basis shall be reduced into laboratory sample on the date immediately following the date of collection.
- f) Samples collected at the loading end shall be analysed as per as per relevant BIS Standards (IS: 1350 Part I) for determination of ash, moisture content and volatile matter and GCV as per (IS: 1350 Part II).
- g) The final pulverized sample will be divided into four equal parts viz. Set – I, Set – II, Set-III and Set-IV. Set – I sample will be retained by Third Party for testing and analysis at their laboratory. Set-II of the sample shall be handed over to the Coal Company. Set-III of the sample shall (i) in case of rail mode, be handed over to the Coal Purchaser. (ii) in case of road mode, be handed over in equal portions to all the Purchasers on that day; and Set-IV of the sample called referee sample is sealed jointly by Third Party and representatives of each of the Parties and is kept at the sample preparation room with lock and key arrangement under custody of Third Party.
- h) The referee sample is retained in double sealed condition (duly signed by the Third Party and the representatives of the Parties) for 30 (thirty) days from the date of sample collection.
- i) Samples drawn at loading ends (Set-I) shall be analysed at designated laboratories of TPS agency.
- j) Samples drawn at loading ends (Set-II & Set-III) shall be analysed in designated NABL accredited laboratories of coal company and power utility respectively.

NOTE 1: At Unloading end the collected samples are divided in three parts.

Set-I- for testing at Unloading end laboratory of power utility.

Set-II – shall be retained by Third party testing and analysis at their laboratory

Set-III shall be treated as referee sample as per the Set-IV samples as described above.

NOTE 2: Each sample shall be assigned with a code number and will be identified by such code only and no other particulars will be indicated or written on the tag attached with the relevant bag containing the sample.

Coal Company/Power Utility shall provide enabling conditions including collection, local transportation of samples to storage room, sample preparation

facilities/machineries and testing equipment's and their storage at loading ends. In addition to this they will be sole responsible for providing infrastructural facilities for collection, i.e., smooth functioning of AMS, ramp/ladders, lighting arrangements, and sample storage room etc. preparation and storage of samples. All the consumables, tools and tackles etc. (standard quality), required for performing the jobs shall be arranged by CSIR-CIMFR/TSP. In the event of any reason whatsoever Third-Party sampling & analysis could not be conducted, then settlement shall be done as per provisions (Clause 4.7.6 (a), (B) & 4.7.7 of FSA.

4.2.2 General Modalities for coal sampling, preparation, and analysis:

Sample shall be collected source wise, grade wise and Power station wise.

- a) Samples shall be collected, packed and transported in such a manner to make these tamper proof.
- b) Name the colliery / siding / Power Station, date of collection and other identification details (eg. Rake no. in case of rail supply) shall be maintained in a register and a proper code number shall be assigned for each sample for identification and reconciliation of results.
- c) Laboratory samples prepared shall be in the size of 12.5mm for Total Moisture and for Proximate Analysis 212 micron IS Sieve. Precaution shall be taken so that before analysis, in test laboratory, further sieving or pulverizing is not required.
- d) Proper analysis records shall be maintained at the laboratories where the samples are analysed. Samples collected at the loading end shall be analysed as per BIS Standards (IS: 1350 Part I) for determination of ash, moisture content and volatile matter and GCV as per (IS: 1350 Part II).
- e) All data to be recorded and maintained in SAP as per the CC-FM guidelines.
- f) The criteria for raising dispute for both loading end samples shall be as per the latest CC-FM guidelines. For Unloading end, criteria for raising disputes shall be as per Bipartite agreement between NTPC and third party. A checklist for enabling conditions at loading and unloading end for coal sampling is to be maintained and submitted to CC-FM on monthly basis.

4.3 COLLECTION OF SAMPLES FROM WAGONS:

- a) NTPC-Simhadri receives coal from rail rakes, each rake (source wise, grade wise and Power Station wise) of Coal supplied from one Delivery Point shall be considered as a Lot for the purpose of sampling.

- b) Each rake shall be divided into sub-lots in a manner that the quantity of Coal/number of wagons in such sub-lots is equal. The number of sub lots shall be determined as per Table-2.

No. of wagons in the rake	Number of Sub lots
Up to 30 wagons	4
>30 wagons up to 50 wagons	5
>50 wagons and above	6

Table – 2

- c) From each of the sub lots one wagon each shall be selected as per random table in IS: 436 (Part I/Section I) or its latest version for collection of increments (Annexure – A : Random tables).
- d) In each wagon selected for sampling, the sample will be drawn from the spot in a manner so that if in one wagon, the sample is collected at one end, in the next wagon the spot will be in the middle of the wagon and in the third wagon at the other end and this sampling procedure will be repeated for subsequent wagons.
- e) Before collecting the samples, the spot will be levelled and at least 25 cm of Coal surface shall be removed/scrapped from the top and the place will be levelled for an area of 50 cm by 50 cm.
- f) About 50 kg of sample shall be collected from each selected wagon in the wake of a source by drawing 10 increments of approx. 5 kg each with the help of shovel/scoop.
- g) Any stone/shale of size more than that as indicated in respective FSA shall be removed/discarded, however all stones/ shale of size in terms of FSA shall form part of the sample collected.
- h) Source wise, grade wise and Power Station Wise Samples collected from all the selected wagons in a rake shall be mixed (grade wise/source wise/Power Station wise) separately to form Gross Sample accordingly.
- i) In case of having live overhead traction line, it is to be ensured that the power supply in the overhead traction is switched off to facilitate collection of joint samples from BOXN / BOBR wagons as mentioned above.
- j) It is recommended not to carry out any sampling work unless the traction wires are made dead and earthed. For NTPC Simhadri, the Isolation procedure and other modalities of live OHE for facilitation in collection of wagon top sampling is enclosed in Annexure-1.

4.4 COLLECTION OF SAMPLES OF COAL DISPATCHES BY ROAD

- a) Sample shall be collected colliery wise / grade wise on daily basis during a day i.e., 6.00 Hr. to 18.00 Hr if safe infrastructural requirements like adequate lighting is not available.
- b) The sampling can be done during 24 hrs also if safe enabling conditions are available for round the clock sampling.
- c) The first truck for joint sampling on a day shall be selected randomly from the first eight trucks placed for loading by the Purchaser. Every eighth (8th) truck there after shall be subjected to joint sampling.
- d) The spot at the top of the truck, will be levelled and at least 25 cm of Coal surface shall be removed/scrapped from the top and the place will be levelled for an area of 50 cm by 50 cm for collection of samples.
- e) About 30 kg of sample shall be collected from each truck by drawing 6 increments of approx. 5 kg each with the help of shovel/scoop.
- f) All the samples collected from every eighth truck shall be mixed grade wise to form a Gross Sample.
- g) Any stone/shale of size more than that as indicated in FSA shall be removed/discarded, however all stone / shale of size as mentioned in as indicated in FSA shall form the part of the sample collected.

4.5 COLLECTION OF SAMPLES FROM STOCKPILE

- a) For the purpose of sampling, the quantity of Coal in the stockpile shall be divided into a suitable manner of sub-lots as specified in the following Table-3 by suitably marking the lines of demarcation on the surface of the lot.

Weight of the lot (MT)	No. of Sub-lots
Up to 500	2
501 to 1000	3
1001 to 2000	4
2001 to 3000	5
Over 3000	6

Table – 3

- b) The surface of each sub-lot shall be levelled and one point for approximately every 250 MT of material in the sub-lots shall be chosen at random for taking gross sample.

- c) In case height of the stockpile is less than 1.5 meter, the material shall be collected at every selected point by taking the whole section of Coal from top to bottom over the area of a circle of 30 cm diameter.
- d) In case the height of the stockpile is more than 1.5 meter, the sample shall be collected at every selected point by taking the material over an area of a circle of 30 cm diameter and up to a depth of 1.5 meter.
- e) The un-used sample after preparation and analysis is to be sent back to coal handling system for re-cycling.

4.6 PREPARATION OF DOMESTIC COAL SAMPLES

Quantity of Sample:

In order to have better representation, the sample should be collected as per procedure in FSA/BIS. The un-used sample after preparation and analysis is to be sent back to coal handling system for re-cycling.

REPARATION OF MANUALLY COLLECTED SAMPLES:

- a) The Gross Sample collected will be divided into two portions. One portion (one fourth of the Gross Sample) called Part-1 will be used for analysis of Total Moisture and the other portion (three fourth of the Gross Sample) called Part-2 for Proximate Analysis and determination of GCV on Equilibrated basis.
- b) The Part-2 Sample shall be reduced into laboratory sample on the date immediately following the date of collection. The Gross Samples shall be separately crushed to (-) 5 cm preferably by mechanical means, mixed thoroughly, coned and quartered.
- c) Two opposite quarters shall be retained, and the rest rejected.
- d) The retained material shall be further mixed, halved and one half retained.
- e) Material so obtained shall be crushed to 12.5 mm by a Jaw Crusher and then to 3.3 mm by a palmac type of reduction mill/ or jaw crusher.
- f) The crushed material shall be reduced either by coning and quartering or by riffing to 2 kgs.
- g) The sample so reduced shall be finally ground to pass through 212 micron IS sieve using a Raymond mini mill.
- h) From the final sample passing through 212 microns IS sieve, 1.5 Kg shall be taken, which shall constitute the laboratory sample.
- i) Such laboratory sample shall be divided into 04 sets for loading and 03 sets for Unloading end respectively. The sample shall be kept in sealed glass or polythene container.
- j) Enabling condition required for collection, preparation, storage and analysis of the sample shall be provided by the CIL or NTPC.
- k) Total Moisture determination will be done at nearest laboratory and remaining tests/analyses (moisture, ash, GCV on equilibrated moisture basis) as per BIS

Standards (IS: 1350 Part I) for determination of ash, moisture content and volatile matter and GCV as per (IS: 1350 Part II) at respective NTPC Simhadri Laboratory/Coal Subsidiary Lab.

4.7 PREPARATION OF TOTAL MOISTURE SAMPLE AND DETERMINATION OF TOTAL MOISTURE:

- a) Part – 1 Sample to be analysed jointly at the Delivery Point for determination of Total Moisture as per IS: 1350(Part –I).
- b) The samples shall be collected and kept in previously weighed airtight plastic containers duly labelled and coded (the weight shall include any sealing material to be used also) immediately.
- c) Weight of such sample shall be minimum 5 kg. The containers shall be sealed at the time of sample collection in such a manner that there is no loss of moisture.
- d) The containers, after the collection of the sample and sealing, shall be individually weighed. Sealing arrangements must be checked thoroughly. All the weights, before and after the collection of samples shall be jointly recorded.
- e) Preparation and analysis of the sample will be done as per SOP ref no. NTPC-SIMHADRI/CHEM/CL/SOP-04.

4.8 MODALITIES FOR IMPORTED COAL QUALITY DETERMINATION

- a) Quality determination of imported coal shall be done at Load port, Discharge port and at NTPC Simhadri.
- b) However, Quality of coal will be final based on the inspection results of the samples taken at NTPC simhadri and payments thereof will be based on such determination at Power stations.
- c) The Successful Bidder shall be responsible for draft survey, sampling and testing of coal samples at Load Port. List of agencies for Load Port is enclosed at **Annexure W of Volume III of the import coal contract agreement.**
- d) The Successful Bidder shall be responsible for draft survey, sampling and testing of coal samples at Discharge Port. List of agencies for Discharge Port/ Transfer Point is enclosed at **Annexure W1 of Volume III of the import coal contract agreement.**
- e) The costs towards draft survey, coal-sampling analysis and testing at various points i.e. Load Port and Discharge Port / Transfer Point shall be borne by the Successful Bidder.
- f) NTPC either directly or through its authorized representative or through an Independent Inspection Agency appointed by it at its discretion and cost can carry out random checking of the sampling & analysis process at Load Port and/or Discharge Port and validate the quantity and quality parameters established by Successful Bidder as above.
- g) **Sampling, Preparation and Testing of coal samples drawn at Load Port/ Discharge Port/ NTPC Simhadri Station end shall be carried out as per Table-4 : (latest versions of ASTM should be considered)**

Parameter	At Load Port/ Discharge Port	At NTPC Power station's end
Sampling		
Sample collection	ASTM D 2234/D 2234M-17	As per sampling Guidelines stipulated in the bidding document
Sample preparation		
Sample preparation	ASTM D-2013	ASTM D-2013
Testing and Analysis of coal samples		
GCV (ARB)	ASTM D 5865-13	ASTM D 5865-13
Total Moisture (ARB)	ASTM D 3173/D 3173 M-17A	ASTM D 3173/D 3173 M-17A
Volatile Matter (ADB)	ASTM D 3175-18	ASTM D 3175-18
Ash (ADB)	ASTM D 3174-12 (2018)	ASTM D 3174-12
Proximate Analysis	ASTM D 3172-13	ASTM D 3172-13
<i>Sulphur (ADB)</i>	<i>ASTM D 4239-18</i>	<i>ASTM D 4239-18</i>
<i>Size</i>	<i>ASTM D 4749-87(2012)</i>	<i>ASTM D 4749-87</i>
<i>HGI</i>	<i>ASTM D 409/D 409M-16</i>	<i>Not Applicable</i>
<i>IDT</i>	<i>ASTM D 1857/D 1857M-18</i>	<i>Not Applicable</i>

Table 4

4.9 IMPORTED COAL - SAMPLING GUIDELINES FOR NTPC SIMHADRI STATION END SAMPLE COLLECTION:

- a) The sample should be representative sample, means it should have coal of all sizes (small, medium, large), all grades (dull, shine) etc. The collected sample should represent the wagon, rake.
- b) The sample quantity of 3-4 Kg each in 4 increments is to be collected from every 3rd wagon of a rake i.e. 12 Kg from each wagon but it should in random.
- c) The randomness can be achieved by following:
- d) With the sample to be collected from every 3rd wagon, one can have 3- set of option in each rake, either start the collecting sample from-1st wagon or 2nd wagon or 3rd wagon. This can be decided by drawing of chit containing the wagon no 1,4,7,10,13----- or 2,5,8,11,14----- or 3,6,9,12,15,18----- on arrival of wagon at NTPC site.
- e) Coal samples at NTPC Simhadri station end, shall be drawn from each rake and prepared for analysis by NTPC in the presence of representatives of HOP, CHP, Chemistry of NTPC and Successful Bidder.
- f) In case the Successful Bidder is not getting involved, while collecting of coal sample and preparing protocol for collection of coal sample, the activity of collection of coal sample should be continued and in such cases the coal sample collected by NTPC or IIAN shall be binding on the Successful Bidder.
- g) After collection of samples from selected wagons, the sample preparation procedure shall be followed as mentioned in subsequent points.
- h) Do the conning of sample, till the sample quantity is reduced to 4-5 Kg.

- i) For Suppose one collects 12 Kg sample [3-4 Kg each in 4 increments] from 18 wagons of a rake, then total coal sample collected will be around 210 Kg per rake.
- j) Prepare a cone (heap) of 210 Kg and divide the cone from top to bottom and remove the half portion (left or right).
- k) Further, prepare the cone for remaining 105 Kg of coal and divide the heap and remove the coal of side opposite to the side of previous removal. Now, the balance available quantity of coal is 52.5 Kg and continue this process till the coal sample is reduced to 4-5 Kg.
- l) Out of this one (1) sample of at least 1 Kg in weight and without grinding shall be taken by NTPC for separate testing.
- m) Also, out of remaining coal sample which has been reduced to 4-5 Kg as above prepare one (1) sample of Coal of adequate quantity without grinding for determination of Total Moisture and Size. For determining the Total Moisture and Size, NTPC shall undertake analysis at its Laboratory immediately (within one hour of sample collection) for Total Moisture and Size. In case of holiday or other than operating hours of the laboratory the samples shall be kept in locker in sealed condition.
- n) Such analysis for determination of Total Moisture and Size by NTPC at its Laboratory shall be carried out by NTPC Chemist in presence of Head of Project (HOP) nominated officer and can be witnessed by Successful Bidder or his representative.
- o) If Successful Bidder wants to take the help of a testing agency for witnessing the analysis of Total Moisture and Size, then he can take the help of agency as per List of Agencies for Discharge Port/ Transfer Point as enclosed at Annexure W1 of Volume III of the Contract document.
- p) The test certificate of Total Moisture and Size shall be signed by NTPC Chemist, HOP nominated officer and Successful Bidder or his representative, immediately after its preparation.
- q) In case the Successful Bidder is not getting involved, while analysing for Total Moisture and Size by NTPC at its Laboratory, the analysis of Total Moisture and Size should be continued and, in such cases, the Total Moisture and Size as determined by NTPC at its Laboratory shall be binding on the Successful Bidder.
- r) No dispute shall be raised subsequently by Successful Bidder or his representative on Total Moisture and Size. NTPC Laboratory operating hours is during the day.
- s) Accordingly, sample(s) for Total Moisture and Size collected beyond NTPC Lab operating hours shall be kept in sealed polythene bag(s) and kept in airtight container(s) under joint lock and joint custody of NTPC and Successful Bidder in NTPC Lab till determination of Total Moisture & Size.
- t) In view of the above, no Referee Sample for determination of Total Moisture and Size shall be kept. Grind the remaining coal sample and sieve it.
- u) Prepare three (3) samples of ground Coal of adequate quantity within one (1) day of its collection and distribute as follows:
 - i. 1st sample –Sample for NTPC for analysis at their end.
 - ii. 2nd sample –Sample to Successful Bidder for testing at his end– this Sample should be taken by Successful Bidder through material exit gate pass from the plant.

- iii. 3rd sample – Referee Sample. Such Referee Sample shall be kept under joint lock and joint custody of Successful Bidder and NTPC, to be used for future reference and/or testing.
- v) NTPC shall analyze its portion of the coal sample, distributed as above, for the balance parameters namely Ash, GCV, Sulphur, Fixed Carbon, Volatile Matter.
- w) Successful Bidder shall analyze his coal, distributed as above, for the balance parameters namely Ash, GCV, Sulphur, Fixed Carbon, Volatile Matter, at the laboratory (ies) as may be identified by him.

5.0 FOR CORRECTIONS ON QUALITY VARIATIONS:

- a) For correction of GCV for landed price, Total Moisture, Ash & Size for quantity and Sulphur for penalty, test results of NTPC shall be considered and in case of any dispute (except for Total Moisture and Size), test results of referee sample, if admissible, shall be considered.
- b) NTPC will submit its test report of each rake to the representative of the Successful Bidder at site. In case of any dispute due to variance in test results of sample collected and distributed (as per Clause 7 (ii) (a) and 7 (ii) (b) of imported coal supply agreement between Successful Bidder and NTPC or IIAN) beyond the limits permissible under the relevant ASTM (**except for Total Moisture and Size**) and subsequent claim by the Successful Bidder for testing of referee sample, the test report [for the sample distributed as Clause 7 (ii) (b)] of corresponding rake of Successful Bidder GUIDELINES ON SAMPLING AND ANALYSIS OF COAL should be provided to NTPC within 10 (Ten) working days after receipt of NTPC report by the Successful Bidder or 20 (Twenty) working days after distribution of sample whichever is later.
- c) Challenge for the testing of referee sample shall be supported by the Bomb Calorimeter print out / screen shot in case of GCV and Printout of Sulphur Analyzer Printout/ Screen shot.
- d) In such an event, Referee Sample shall be forwarded by NTPC to Third Party Inspection Agency nominated by NTPC i.e. Central Institute of Mining and Fuel Research (CIMFR). The cost of testing of the Referee Sample at CIMFR shall be borne by the Successful Bidder. NTPC shall not be responsible for the time taken for Referee Sample tested by CIMFR.
- e) If no dispute is raised by Successful Bidder within Ten (10) working days after receipt of NTPC report by the Successful Bidder or twenty (20) working days after distribution of sample whichever is later, no request for consideration of referee sample shall be accepted subsequently. In such cases, analysis report of NTPC shall be binding and NTPC reserves the right to destroy the Referee Samples.
- f) While forwarding the referee sample, Station must ensure the confidentiality of sample to be tested at various CIMFR labs. Only on receipt of CIMFR result of referee sample, the same shall be communicated to Successful Bidder for resolution of dispute.

5.1 GROSS CALORIFIC VALUE BASED GRADING SYSTEM

Pursuant to the decision of the Govt. of India to switch over from existing Useful Heat Value (UHV) based system of grading and pricing of Non-Coking Coal of the country to fully Gross Calorific Value (GCV) based system w.e.f. 01.10.2012, different GCV bands has been identified for pricing of coal are as mentioned below table - 5:

GCV BAND	RANGE	GCV BAND	RANGE
G1	> 7000	G10	> 4300 - 4600
G2	> 6700 - 7000	G11	> 4000 - 4300
G3	> 6400 - 6700	G12	> 3700 - 4000
G4	> 6100 - 6400	G13	> 3400 - 3700
G5	> 5800 - 6100	G14	> 3100 - 3400
G6	> 5500 - 5800	G15	> 2800 - 3100
G7	> 5200 - 5500	G16	> 2500 - 2800
G8	> 4900 - 5200	G17	> 2200 - 2500
G9		> 4600 - 4900	

Table- 5

5.2 SUMMARY OF TEST METHODS USED FOR COAL SAMPLES ANALYSIS

The following test methods (Table -6) shall be used for testing of coal Sample's analysis: (For Imported Coal, Sampling is to be done as per ASTM D-2013 and for domestic Coal sampling methodology will be as per Tripartite/Bilateral agreement for Unloading and loading end respectively)

S.No.	PARAMETERS	STANDARD (LATEST Version)
1	Inherent Moisture (IM)	ASTM D 7582
2	Air-Dried moisture (ADM)	IS: 1350 (Part I) -1984, Reaffirmed-2019
3	Equilibrated Moisture (EM)	IS: 1350 (Part I) -1984, Reaffirmed-2019
4	Total Moisture	IS: 1350 (Part I) -1984, Reaffirmed-2019, ASTM D-3302
4	Ash (A)	ASTM D 7582 IS: 1350 (Part I) -1984, Reaffirmed-2019
5	Volatile Matter (VM)	ASTM D 7582 IS: 1350 (Part I) -1984, Reaffirmed-2019
6	Fixed Carbon (FC)	ASTM D 7582 & IS: 1350 (Part I) -1984, Reaffirmed-2019
7	Sulphur (S)	ASTM D 4239-18e1, IS :1350 PART-3
8	Ultimate Analysis	ASTM D-3176, IS :1350, PART-4 SECTION-1
8.	NITROGEN (N)	IS :1350 PART-4-SECTION-2, ASTM D-3179
9	HGI	ASTM D409,IS:4433
10	IDT	ASTM D 1857/D 1857M-18
11	Gross Calorific Value (GCV)	ASTM D 5865/5865M-19 IS: 1350 (Part 2) -2017 [ISO 1928:2009]

Table- 6

5.3 REPORTING OF RESULTS

The results of proximate analysis and calorific value may be reported in the following manner, making use of the recommended symbols.

5.3.1 Air - Dried Basis

Proximate analysis and calorific value of coal shall be determined on the air-dried laboratory sample of coal.

The following symbols shall be used without any suffix.

M = Percentage of moisture in the sample

A = Percentage of ash in the sample

V = Percentage of volatile matter in the sample

F.C = Percentage of fixed carbon in the sample

Q = Calorific value of the sample

S = Percentage of total Sulphur in the sample.

It is to be noted that the volatile matter V, is the total loss recorded minus the moisture given off on drying (at $108 \pm 2^\circ\text{C}$).

Fixed carbon F.C is calculated as $= 100 - (M + A + V)$

5.3.2 Received Basis

The symbols as indicated 5.2.1 shall be used with suffix 1. The results on the as received basis are then obtained as follows:

At Moisture = M1

For example, the Ash (%) as received will be calculated as $A1 = A \times (100 - M1) / (100 - M)$

In the similar way V1, F1, Q1 and S1 may be determined by multiplying $(100 - M1) / (100 - M)$ with V, F, Q and S respectively.

5.3.3 Air-Dry Basis for equilibrated condition:

(Sample Equilibrated at 40°C and 60% Relative Humidity). The symbols indicated in 5.2.1 may be used with suffix 2.

The results on the air-dry basis are then obtained as follows (for ash):

$A2 = A \times (100 - M2) / (100 - M)$

In the same way V2, F2, Q2 and S2 can be obtained by multiplying $(100 - M2) / (100 - M)$ with V, F, Q and S respectively.

5.3.4 Dry Basis

The symbols indicated in 4.10.1 shall be used with suffix d.

The results on dry basis are then obtained as follows: -

$Ad = A \times 100 / (100 - M)$

In the similar way Vd, Fd, Qd and Sd can be determined, by multiplying 100 with V, F, Q, and S respectively.

5.3.5 Dry Mineral Matter Free Basis

The symbols indicated in (1) shall be used with suffix dmf. The results on the dry mineral matter free basis are then obtained as follows –

$F.C_{dmf} = F.C \times 100 / 100 - (M + 1.1A)$

$V_{dmf} = 100 - F.C_{dmf}$

The following points may be noted while making calculation on dmf basis.

The above calculation assumes that mineral matter is equal to 1.1 Ash and sulphur less than 1% and CO₂ less than 0.5%.

In obtaining Sdmf allowance shall be made for 5% as FeS₂ and sulphates. It is only the organic Sulphur, which can be expressed on a dmf basis.

B) In obtaining Cdmf, it is necessary again to ensure that in the total 'C' figure given from ultimate analysis, the C present as carbonate has been deducted.

5.4 CRITERIA FOR REJECTION OF SUSPECT TEST RESULTS

It may sometime happen that a test result is obtained which deviates considerably from the other test results and therefore arouses suspicion that it may have arisen from a mistake in sampling or sample preparation, rather than the course of normal variation. In the case that the chemist has confirmed a mistake has occurred during analysis, the test result must be rejected irrespective of its magnitude. If, however, only a suspicion exists, it may be desirable to determine whether such a result may be rejected or whether it must be accepted as part of the normal variation expected.

5.4.1-Criteria

The following procedure may be followed for determining the acceptability or rejectability of the suspect test results :

Arrange all the test results obtained by analysing 'n' laboratory samples in the ascending order of magnitude and designate them X₁, X₂,....., X_n, where X₁ is the smallest and X_n the largest test results.

If X_n is suspect, calculate the value of K₁ from $K_1 = (X_n - X_{n-1})/(X_n - X_1)$

If X₁ is suspect, calculate the value of K₂ from $K_2 = (X_2 - X_{n-1})/(X_n - X_1)$

If the calculated value of K₁ or K₂ exceeds the corresponding value given in the following Table, the suspect result shall be rejected. The above procedure can also be repeated to remove the suspected test results other than the first removed.

5.4.2 - VALUES OF CRITERION K₁ OR K₂

Sample size (n)	Value of K ₁ or K ₂
3	0.941
4	0.765
5	0.642
6	0.560

Table- 7

Example:

In an investigation on ash percentage of coal, the following test results are obtained. 28.9, 26.9, 26.8, 26.7, 26.1

It is required to find out whether 28.9, which deviates rather unduly from the others could be considered as suspect.

The test results when arranged in ascending order, are:

26.1, 26.7, 26.8, 26.9 and 28.9

According to the criterion given in

$$K1 = (28.9 - 26.9) / (28.9 - 26.1) = 2.0 / 2.8 = 0.714$$

Since this value of K1 exceeds the corresponding tabulated value 0.642 for sample size 5, the test result shall be rejected.

5.5 ASSOCIATED DOCUMENTS- (References)

- a) *All latest versions of BIS and ASTM standards.*
- b) *Guidelines for isolation of OHE for wagon top sampling at Track hopper Annexure -1.*

Annexure – 1

PROCEDURE OF WAGON TOP SAMPLING IN TRACK HOPPERS-OHE ACTIVITIES

- 1) On passing of the BOBRN load rake from Crossing station, the two persons nominated for operation of Isolators to be sent to the isolator locations (Either end of TH-1 or TH2, as per CHP requirement).

Responsibility : Track Hopper Shift In charge

- 2) On arrival of the load rake at TH, unload the starting 04 wagons first & then move the rake & position the fifth wagon, so that loco remains outside the front isolator zone. Shift in-charge should confirm that the engine has passed the Prescribed Sign boards for loco position.

Responsibility : Track Hopper Shift In charge, Shift supervisor.

- 3) Operate the isolator 249 (OHE post no 26/25) for TH-1 and 248 (OHE post no 26L.12/21A) for TH-2, which is at the rear of the Track hopper first. After this isolator is operated, then operate the isolator 249A (OHE post no 26/17) for TH-1 and 248A (OHE post no 26L17/15A) for TH-2, near the TH control room. (The communication between the two persons nominated for isolator operation shall be through walkie talkies)-With this operation(when both isolators are operated-the LED bulbs available in TH, would change from RED to GREEN-An indication that OHE has been put off)

*Caution-Never operate the isolator near the TH control room first.

- Caution- loco should not be shunted back.

Responsibility : Track Hopper Shift In charge

- 4) Lock both the isolators & the keys of the same to be handed over to shift in charge. The sampling staff shall then climb to wagon top for sampling, after taking clearance from Track hopper shift In charge & on observing GREEN LED lamps in Track hopper area.

Responsibility : CSIR-CIMFR, COTECNA, Track Hopper Shift Incharge

- 5) Unloading of the wagons & sampling would now be carried out simultaneously-**Caution - Unloading staff to ensure that no person is on the wagon top, whose doors are being opened**

Responsibility : Shift supervisor.

- 6) After completion of unloading of all the wagons, the empty rake would be departed to R&D yard

Responsibility : Track Hopper Shift In charge

- 7) After completion of the last wagon of the rake, person to be deployed to the isolator 249A(OHE post no 26/17) for TH-1 and 248A (OHE post no 26L17/15A) for TH-2 at the TH control room, Only After the receipt of clear communication from Shift In-charge, which is to be given only after the last wagon has left the track hopper area.

Responsibility : Track Hopper Shift In charge

- 8) After that the isolator 249 (OHE post no 26/25) for TH-1 and 248 (OHE post no 26L.12/21A) for TH-2, at the rear of the Track hopper is to be operated, thereby causing the OHE to be charged & LED bulbs changing to RED.

Responsibility : Track Hopper Shift In charge

- 9) Note: No piece meal wagon placement, inside the track hoppers would be done in the first placement, when loco has been positioned near the track hopper control room isolator & OHE is put off within the track hopper zone.
BOTH ISOLATOR KEYS SHOULD BE IN THE CUSTODY OF SHIFT IN CHARGES ALL THE TIME.
- 10) After unloading completion, before movement of empty BOBR from track hopper, Traffic supervisor will blow hooter.
- 11) TH to ensure that OHE isolators at both sides of Track Hopper and earthing system are healthy.
- 12) During unloading of wagons, Unloading Supervisor shall take utmost care that no one climbs on the wagon. In case of wet & sticky coal, looking the wagons from the wagon's bottom gate shall be done with safe distance from bottom gates.
- 13) Unloading Supervisor will always be present with his workers and will ensure their safety; he will also ensure the use of PPE by himself and his workers.
- 14) All activities as mentioned as mentioned in SL no- (1) to (13) shall be recorded and documents for the same is to be maintained.
- 15) In case of diesel locomotive in BOBR and track hopper being electrified any sampling should be taken up only after ensuring isolation of OHE as per written memo system explained above.
- 16) At Wagon tippler Sampling platform must be located at position where 25KV OHE is not there.



Thank you!